ence and Technology, CS Department
Second Exam, CS451
Fall 2025/2026

164425

Q1: Select the most correct answer: (15 points)

1. What's the equivalent of \$t2=A[A[0]] in assembly where the base address of A is in \$s0:

a.
```
lw $t0, 0($s0)
lw $t2, 0($t0)
```
b.
```
lw $t0, 0($s0)
sll $t1, $t0, 2
add $t1, $s0, $t1
lw $t2, 0($t1)
```
c.
```
lw $t0, 0($s0)
add $t1, $s0, $t0
lw $t2, 0($t1)
```
d.
```
lw $t2, 0($s0)
```

2. Which of the following instructions uses PC-relative addressing?
   a. addi \$t1, \$t2, 10
   b. lw \$t0, 4(\$s0)
   c. jr \$ra
   d. beq \$t0, \$t1, LABEL

3. Assume a new instruction **move \$t0, \$t1** that copies the contents of **\$t1 to \$t0**. *Which of the following standard MIPS instructions is **NOT** equivalent to this operation?*
   a. add \$t0, \$t1, \$zero
   b. addi \$t0, \$t1, 0
   c. sub \$t0, \$t1, \$zero
   d. sub \$t0, \$zero, \$t1

4. Decode the machine instruction **0x014B4820** :
   a. add \$t1, \$t2, \$t3
   b. addi \$t1, \$t2, 20
   c. and \$t1, \$t2, \$t2
   d. sub \$t1, \$t2, \$t3

5. Which of the following registers is used with one of J-type instructions?
   a. \$t0
   b. \$ra
   c. \$s0
   d. \$s3

6. What's the final value stored in \$t0 after executing **lui \$t0, 0x1234,** assuming \$t0 originally contains 0?
   a. 0x12340000
   b. 0x00001234
   c. 0xFFFF1234
   d. 0x1234FFFF

7. If a **beq \$t0, \$t1, Label** at address 0x00400020 branches to 0x00400028, what is the value of **Label**?
   a. 1
   b. 6
   c. 4
   d. 8

Student name:

Student ID:

8. In which case can signed overflow occur in the operation A – B :
   a. A is positive and B is positive
   b. A = 0 and B = 0
   c. A is positive and B is negative
   d. A is negative and B is negative

9. Assume a 4-bits multiplicand and a 4-bits multiplier, what's the size of the ALU that could perform the first-version multiplication :
   a. 8-bits ALU
   b. 16-bits ALU
   c. 4-bits ALU
   d. 32-bits ALU

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10. What is the IEEE 754 single-precision representation of the decimal value −1.5?
   a. 1011 1111 0110 0000 0000 0000 0000 0000
   b. 0100 0000 0100 0000 0000 0000 0000 0000
   c. 1000 0000 0110 0000 0000 0000 0000 0000
   d. 1011 1111 1100 0000 0000 0000 0000 0000

10.5

Write your choice in the following Answers table. Other answers located outside the table will not be considered.

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
a	c	d	a	b	d	a	c	a	d

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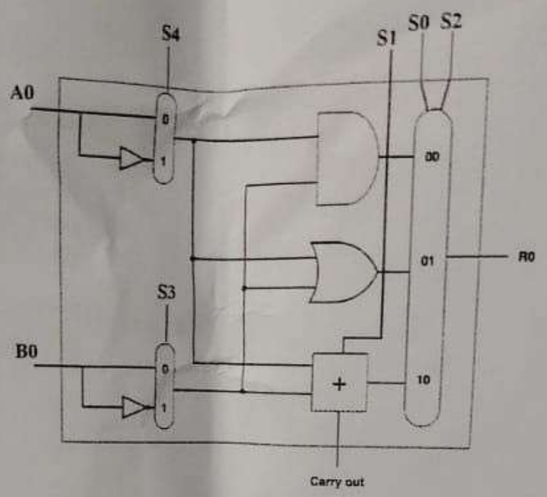
Q2: Based on Chapter 3 (ALU Design). Your goal is to understand the following design and answer the questions associated to it. **Consider the distribution of the selectors** (15 points)



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1. [2.5 points] Write down input signals to perform subtraction (A – B):

S4	S3	S2	S1	S0
0	1	0	1	1

2. [3.5 points] Write down input signals to support a new instruction called **"INC A"** which increments **A** by 1 (A = A+1):

A0	B0	S4	S3	S2	S1	S0
A	1	0	0	0	0	1

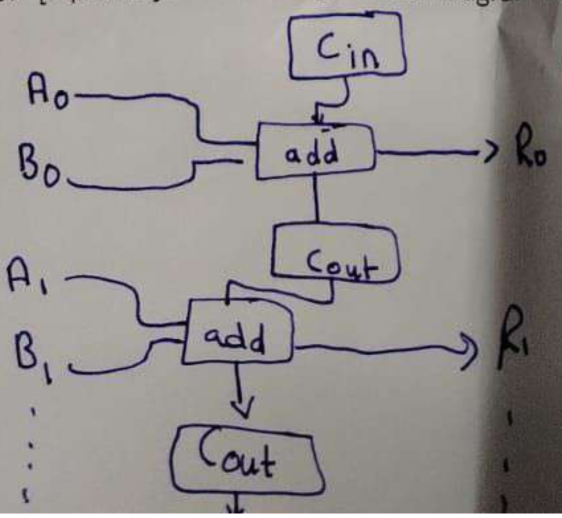
3. [3.5 points] Write down input signals to support a new instruction called **"DEC A"** which decrements **A** by 1 (A= A-1):

A0	B0	S4	S3	S2	S1	S0
A	1	0	1	0	1	1

4. [3.5 points] Write down input signals to support a new instruction called **"INV A"** which inverts **A**. For example, if A=0001 (0111), then **INV A** returns 1000:

A0	B0	S4	S3	S2	S1	S0
A	1	1	0	0	0	0
A	0	1	0	1	0	0

5. [2 points] Draw the logic circuit diagram for the **carryout** in the full adder:



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